

MAT627 Chapter 4 Programming Assignment

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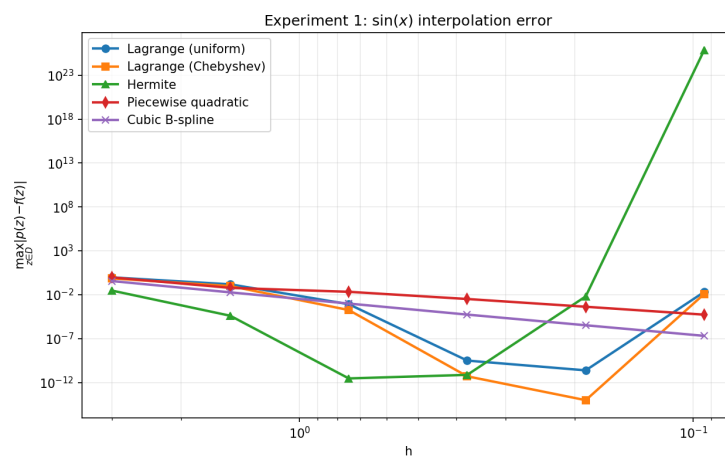


Figure 1: Experiment 1: maximum interpolation error for $\sin(x)$ on $[-3, 3]$ across the five methods.

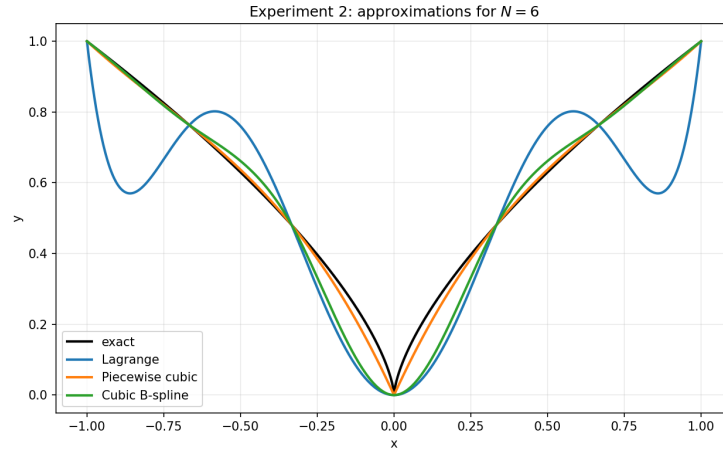


Figure 2: Experiment 2: approximations for $f(x) = \sqrt[3]{x^2}$ on $[-1, 1]$ with $N = 6$.

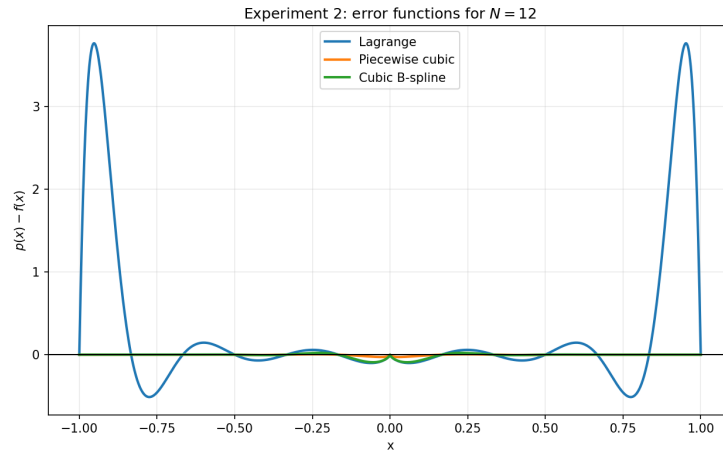


Figure 3: Experiment 2: error functions for $N = 12$.

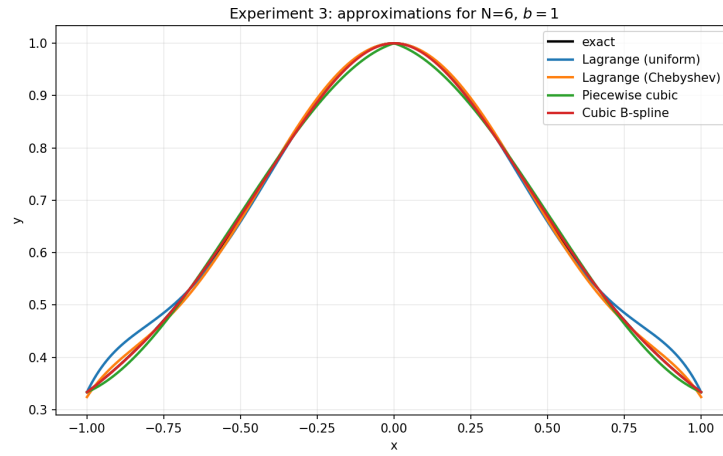


Figure 4: Experiment 3 with $b = 1$: approximations for $N = 6$.

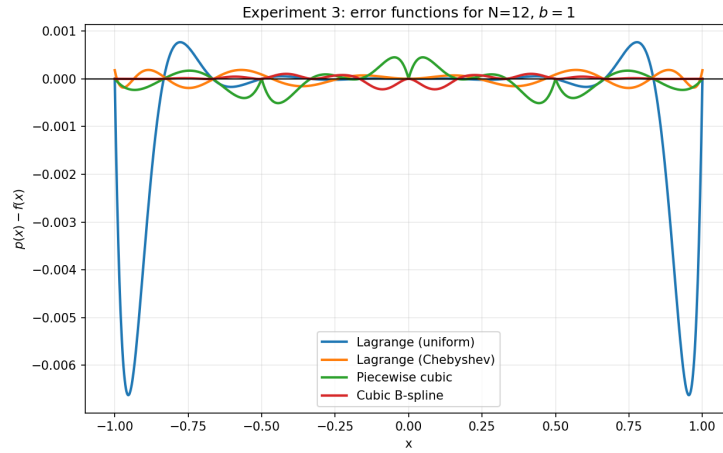


Figure 5: Experiment 3 with $b = 1$: error functions for $N = 12$.

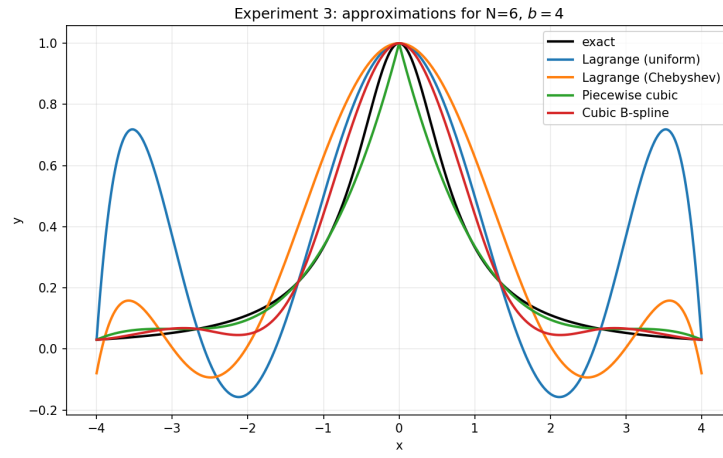


Figure 6: Experiment 3 with $b = 4$: approximations for $N = 6$.

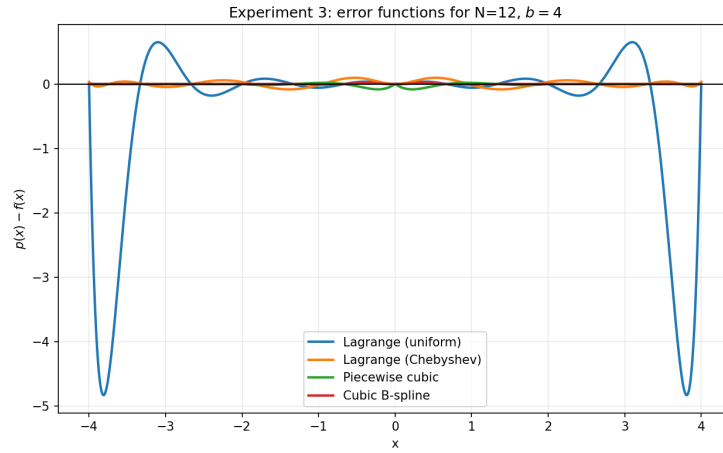


Figure 7: Experiment 3 with $b = 4$: error functions for $N = 12$.

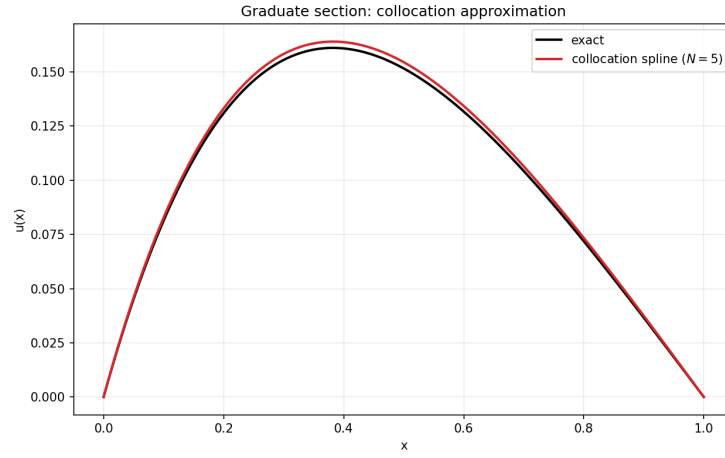


Figure 8: Graduate section: collocation spline approximation versus the exact solution for $N = 5$.

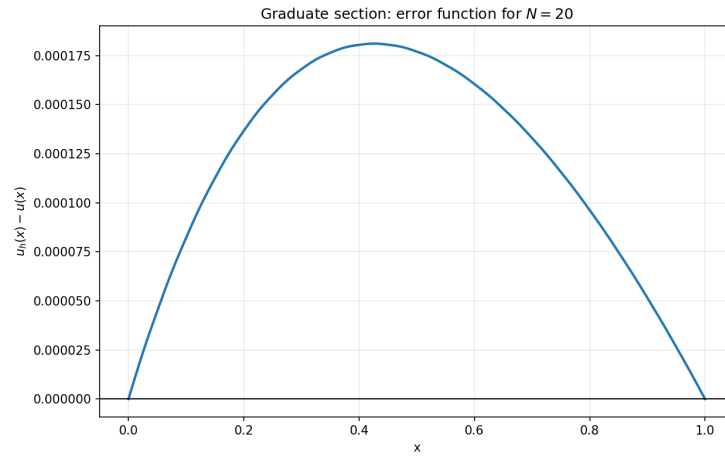


Figure 9: Graduate section: error function $u_h - u$ for $N = 20$.

1 Experiment 1

Table 1: Maximum fine-grid interpolation errors for experiment 1.

N	Lagrange 1	Lagrange 2	Hermite	Piecewise Quadratic	Cubic Splines
2	9.272163×10^{-1}	7.072058×10^{-1}	2.921950×10^{-2}	9.272163×10^{-1}	3.549445×10^{-1}
4	1.498450×10^{-1}	9.443894×10^{-2}	3.934819×10^{-5}	5.912027×10^{-2}	1.816593×10^{-2}
8	8.137846×10^{-4}	1.756125×10^{-4}	2.838785×10^{-12}	2.103657×10^{-2}	9.630480×10^{-4}
16	3.069895×10^{-10}	4.951317×10^{-12}	7.226941×10^{-12}	3.185028×10^{-3}	5.307879×10^{-5}
32	2.358799×10^{-11}	9.409140×10^{-15}	6.297848×10^{-3}	4.165453×10^{-4}	3.242610×10^{-6}
64	2.018284×10^{-2}	1.192288×10^{-2}	7.314523×10^{25}	5.250965×10^{-5}	2.013138×10^{-7}

2 Experiment 2

Table 2: Maximum fine-grid interpolation errors for experiment 2.

N	Lagrange	Piecewise Cubic	Cubic Splines
6	3.425565×10^{-1}	4.524767×10^{-2}	1.451182×10^{-1}
12	3.765897×10^0	2.850324×10^{-2}	9.144778×10^{-2}
24	2.790781×10^3	1.794610×10^{-2}	5.760850×10^{-2}
48	8.781454×10^9	1.129146×10^{-2}	3.629108×10^{-2}

3 Experiment 3 (b=1)

Table 3: Maximum fine-grid interpolation errors for experiment 3 (b=1).

N	Lagrange 1	Lagrange 2	Piecewise Cubic	Cubic Splines
6	3.485296×10^{-2}	9.883290×10^{-3}	1.280884×10^{-2}	2.754402×10^{-3}
12	6.624231×10^{-3}	1.912640×10^{-4}	5.088360×10^{-4}	2.192263×10^{-4}
24	4.266948×10^{-4}	7.082722×10^{-8}	1.304433×10^{-4}	1.265288×10^{-5}
48	1.660334×10^{-5}	1.174116×10^{-10}	1.095460×10^{-5}	7.601237×10^{-7}

Table 4: Maximum fine-grid interpolation errors for experiment 3 (b=4).

N	Lagrange 1	Lagrange 2	Piecewise Cubic	Cubic Splines
6	6.798086×10^{-1}	3.133067×10^{-1}	1.063409×10^{-1}	1.692024×10^{-1}
12	4.829506×10^0	9.736401×10^{-2}	8.231781×10^{-2}	1.599869×10^{-2}
24	4.982443×10^2	1.229606×10^{-2}	1.280106×10^{-2}	2.785866×10^{-3}
48	9.332426×10^6	1.216318×10^{-3}	5.071824×10^{-4}	2.191979×10^{-4}

4 Experiment 3 (b=4)

5 Graduate Section

Table 5: Fine-grid collocation errors for $-u'' + u = 4e^{-x} - 4xe^{-x}$ with exact solution $u(x) = x(1-x)e^{-x}$.

N	h	$E_h = \max_{z \in D} u_h(z) - u(z) $	Rate
5	0.2	2.924971×10^{-3}	–
10	0.1	7.262757×10^{-4}	2.009833
20	0.05	1.810827×10^{-4}	2.003868
40	0.025	4.522158×10^{-5}	2.001566
80	0.0125	1.130428×10^{-5}	2.000143
160	0.00625	2.825992×10^{-6}	2.000040

6 Discussion

Experiment 1

For the smooth sine function, the local methods are the most reliable. The piecewise quadratic interpolant and the cubic spline converge cleanly, with the spline showing fourth-order convergence. The Hermite and global Lagrange formulas become extremely accurate at moderate N , but by the largest values they also show the finite-precision instability expected from high-degree global divided-difference polynomials. Among the global Lagrange methods, the Chebyshev nodes are more stable than the equally spaced nodes.

Experiment 2

The cusp at $x = 0$ causes problems for all methods. The global Lagrange interpolant with equally spaced nodes becomes rapidly unstable as N grows, while the piecewise cubic and spline approximations continue to improve with a much more modest rate. That slower decay is still consistent with the limited smoothness of $f(x) = \sqrt[3]{x^2}$: the local methods are stable, but the singular derivative prevents the high-order convergence seen in the smooth experiments.

Experiment 3

When $b = 1$, every method is reasonably accurate, but the Chebyshev-node interpolant already has the smallest global oscillation. When $b = 4$, the equally spaced global Lagrange interpolant develops the large endpoint oscillations as described by the Runge phenomenon, while the Chebyshev, piecewise cubic, and spline approximations stay substantially more stable.

Graduate Section

The collocation spline produces a smooth approximation on the full interval and the error table shows second-order decay. Because the approximation is represented directly as a cubic spline, the boundary value problem is solved over the whole interval rather than only at the mesh points, so the plotted error is smooth instead of like with a finite difference method.